Audio Filtering Pipeline for Indic Speech

Assignment Overview:

Implemented a scalable, automated curation pipeline that is designed to filter out bad quality audio from the **IndicVoices** dataset. It processes raw audio in three primary stages:

1. **Data Ingestion:** Efficiently downloading, parsing, and extracting raw audio bytes directly from Hugging Face parquet files into structured .flac directories.
2. **Context-Aware Acoustic Filtration:** Evaluating audio quality using a custom, per-language normalized "Soft Score" to surgically remove noise, dead air, and distortion without introducing linguistic bias.
3. **Semantic Alignment & Verification:** Used multi-stage **faster-whisper** pipeline to compare audio against transcripts given in dataset and drop severe transcript mismatches.

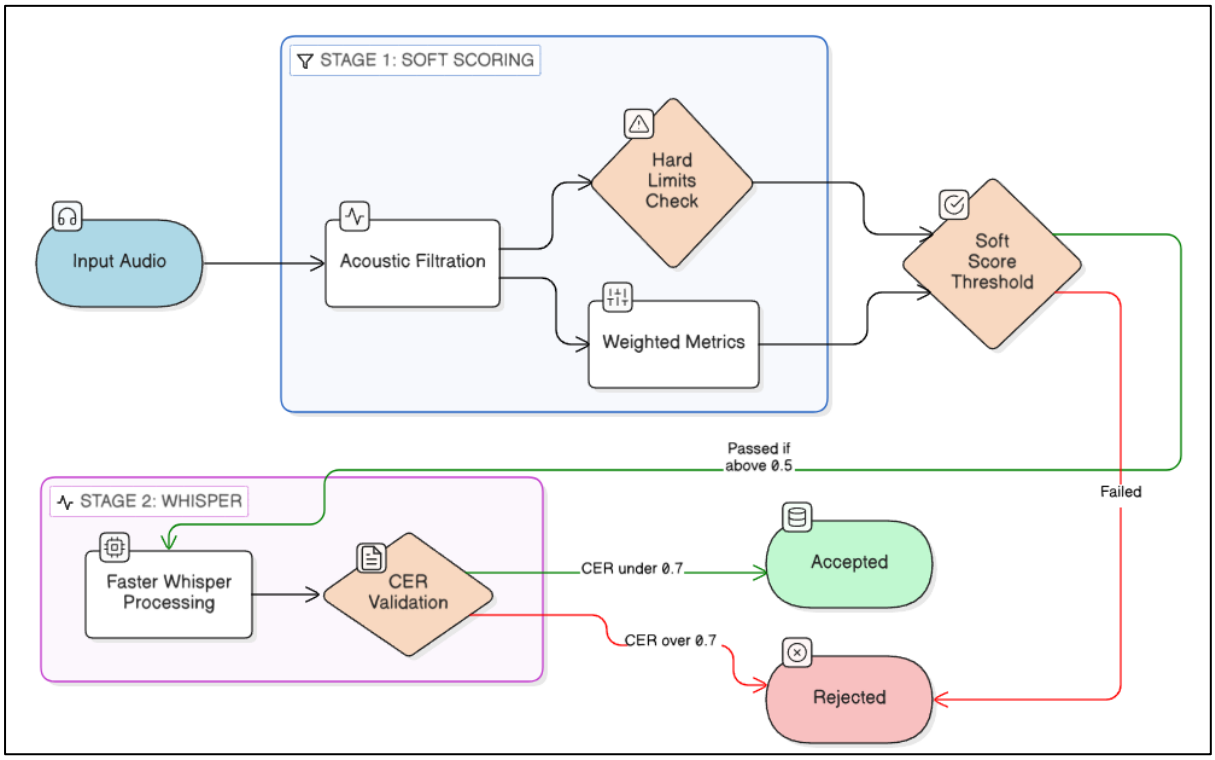
Dataset:

- **Source:** IndicVoices by AI4Bharat
- **Languages Evaluated:** Hindi, Bengali, Marathi, Gujarati, Tamil, Telugu, Kannada, Malayalam
- **Audio Format:** Extracted and standardized to 16 kHz .flac Mono
- **Metadata:** Managed via structured .jsonl manifests containing original text, normalized text, and demographics

Key Engineering Features:

- **Dynamic Hardware Profiling:** Auto-detects available CPU cores and GPU VRAM to scale batch sizes and compute types (float16 vs. int8), preventing OOM crashes.
- **"Zero-Idle" Semantic Pipeline:** Custom 3-stage concurrent architecture (I/O Prefetching → GPU Transcription → CPU Math) ensuring the GPU is never stalled.
- **Hybrid Normalization Soft-Score:** Balances 7 distinct audio metrics using per-language normalization for variable metrics (VAD Ratio, SNR, ZCR) and global normalization for universal structural flaws.
- **Acoustic Profiling Suite:** Automatically generates diagnostic visualizations — Boxplots, Histograms, Radar Charts, and Retention Curves — to mathematically validate rejection thresholds.

Architectural Pipeline:



Soft Scoring (Core Filtration):

The Soft Scoring pipeline is the core of this assignment. It computes a composite audio quality score for each file by evaluating 7 acoustic metrics, normalizing them intelligently (per-language vs. global), and combining them with tuned weights.

The 7 Acoustic Metrics:

Metric	What It Detects	Target Flaw
SNR (WADA)	Constant background noise (AC hum, traffic)	Prevents model hallucinations from noise floor
VAD Ratio	Percentage of actual human speech in recording	Dead air, silence-heavy files
ELR (c50_db)	Room echo and reverberation level	Smeared word boundaries, echoey rooms
Clipping Rate	Digital distortion from too-loud recording	Blown-out files with destroyed speech data
Waveform Kurtosis	Sudden extreme amplitude spikes	Mic pops, table bumps, impulse noise
Spectral Flatness	Ratio of structured speech vs. flat noise	Files dominated by white noise or drones

ZCR	Sound wave vibration frequency	Electronic hissing, high-pitched whines
Silence Ratio	Percentage of absolute dead air	Recordings with long, unnatural pauses

Key Inferences from Acoustic Visualizations:

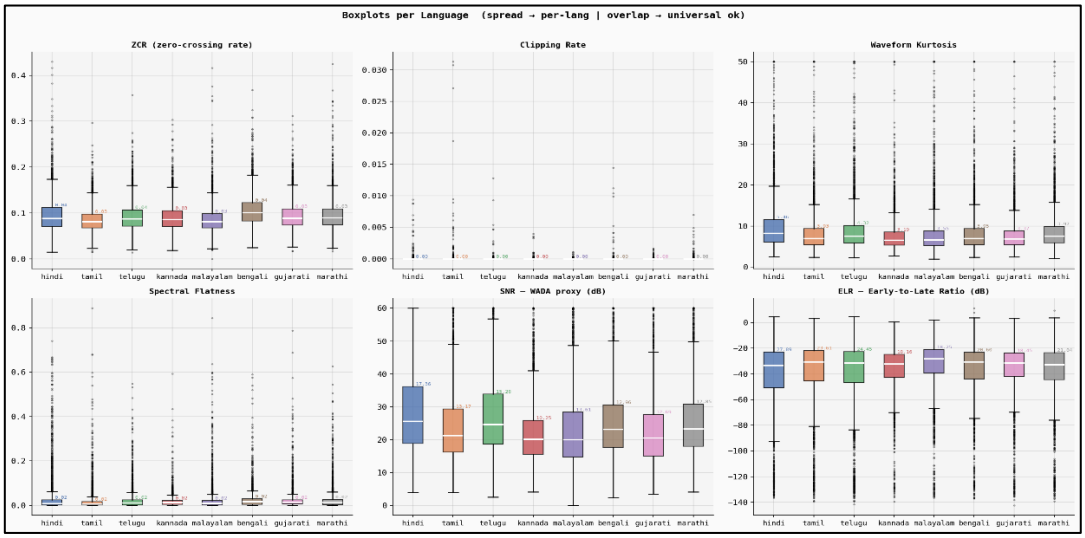
A. Global vs. Per-Language Normalization: Kurtosis, Spectral Flatness, and ELR are stable across languages **(CV < 0.25)** → global norm. VAD Ratio and SNR vary wildly — a global VAD threshold less than **0.60 deletes 40%** of Hindi while keeping all Tamil → per-language norm mandatory.

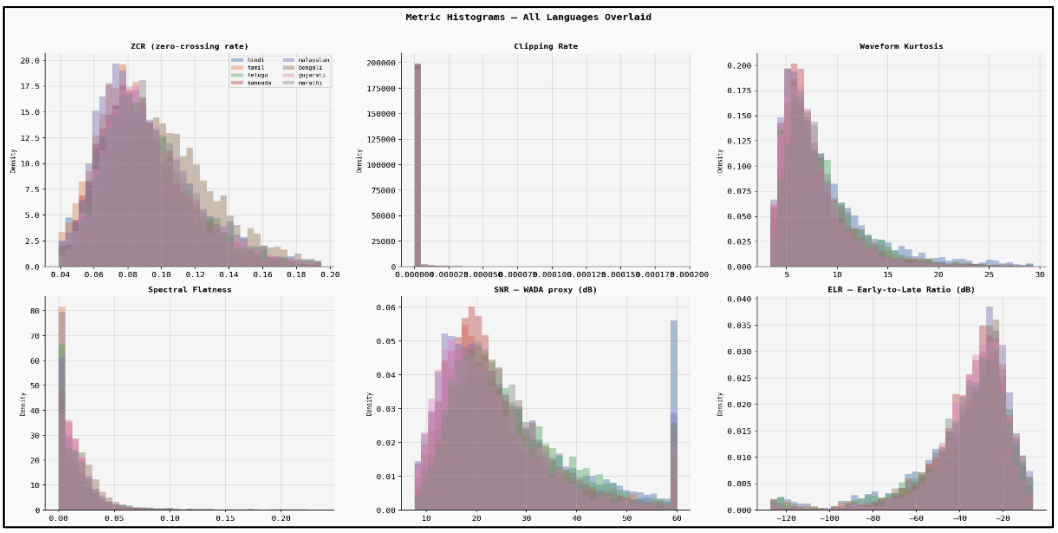
B. Redundant Metrics & Hard Failures: VAD Ratio and Silence Ratio are near-perfectly inversely correlated **(r = –0.91)** → Silence Ratio dropped. Clipping Rate has near-zero correlation with acoustic metrics (hardware error) → reclassified as binary Hard Reject.

C. Language-Specific Quirks: Bengali has anomalously high ZCR — a linguistic trait, not an audio defect. Decision: ZCR moved to per-language normalization so Bengali is benchmarked against its own baseline only.

Visualiation:

Box plot and Histogram per language for all metrics





The Three-Step Curation Pipeline

Step 1: Hard Filters (Immediate Rejection)

Applied before any scoring. Files are immediately rejected if:

- **Duration < 0.3 seconds** (too short to contain valid speech)
- **Clipping Rate > 0.05** (heavy permanent digital distortion)
- **VAD Ratio < lang_median_VAD × 0.40** (essentially dead air)

Step 2: Hybrid Normalization (0.0 → 1.0)

All surviving metrics are normalized to a common [0.0, 1.0] scale to allow combining different units (dB, percentages, raw floats):

- **Per-Language Normalization:** Applied to VAD Ratio, SNR, and ZCR — metrics with high natural variance across languages **(CV > 0.25).**
- **Global Normalization:** Applied to ELR (c50_db), Spectral Flatness, and Kurtosis — structurally stable metrics (CV < 0.25) where a reverberant room sounds equally degraded in any language.

Step 3: Weighted Soft Score — Initial vs. Tuned Configuration

Initial weight configuration (before tuning):

Metric	Weight	Normalization
SNR (WADA)	40%	Per-Language
VAD Ratio	30%	Per-Language
ELR (c50_db)	10%	Global
Spectral Flatness	10%	Global
ZCR	5%	Per-Language

Kurtosis	5%	Global

`SOFT_SCORE_THRESHOLD = 0.40`

The initial 0.40 threshold yielded only ~**2.2% rejection** — too low given manual audits indicating ~**10%** of samples had quality issues. After CDF analysis, the threshold was moved to **0.50 (the natural quality cliff),** raising rejection to ~**5%** and correcting the miscalibration.

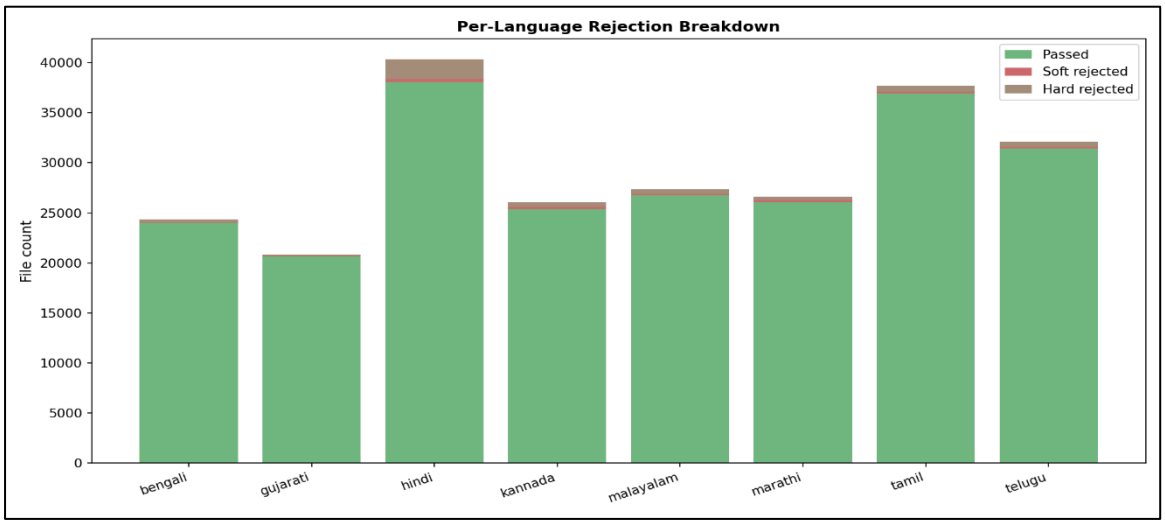
After Tuning:

Metric	Weight	Normalization
SNR (WADA)	**45%**	Per-Language
VAD Ratio	**30%**	Per-Language
ELR (c50_db)	**10%**	Global
Spectral Flatness	**5%**	Global
ZCR	**5%**	Per-Language
Kurtosis	**5%**	Global

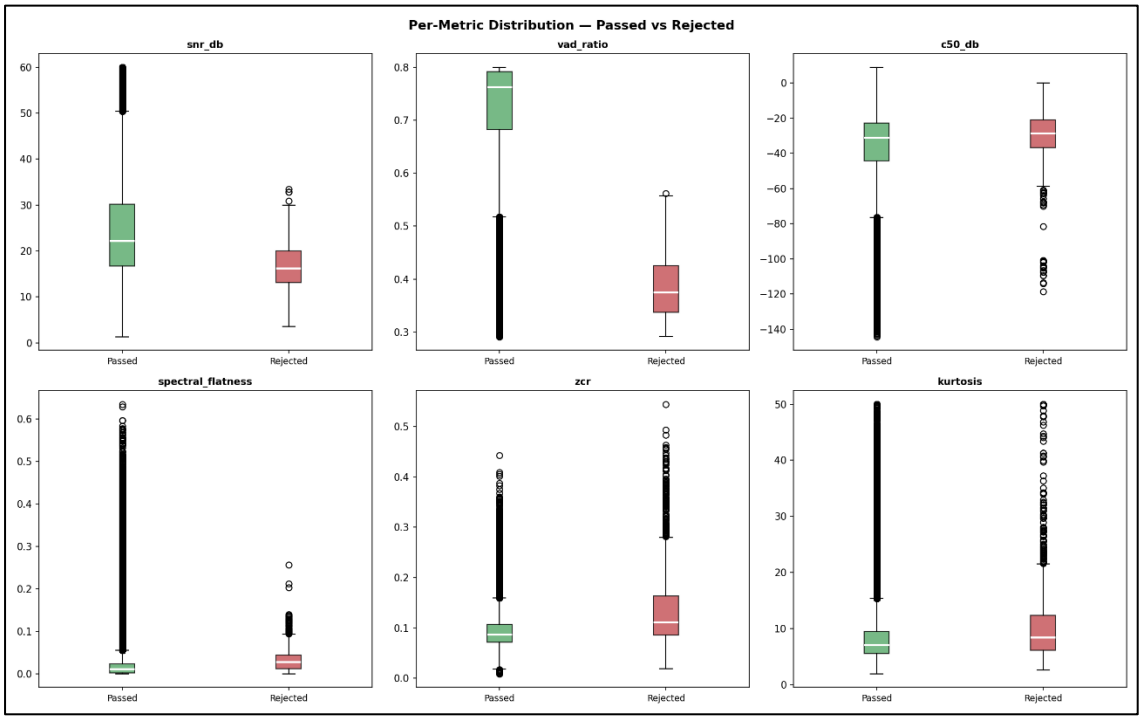
`SOFT_SCORE_THRESHOLD = 0.50`

Plots deriving Insights:

Language wise rejection:



Per-Metric Distribution– Passed vs Rejected Auido samples:



Semantic Alignment via Whisper (Bonus):

- Previous phase filtered acoustic flaws but not semantic alignment — a studio-quality file can still have the wrong speaker, hallucinated speech, or a completely mismatched transcript.
- This Phase resolves this using **Whisper + CER** — each audio file from filtered_manifest.json is transcribed and scored against its ground-truth text to accept or reject the sample.
- Model performance is moderate since faster-whisper **wasn't optimized for Indic languages.**
- Pipeline: ***Audio File → Whisper Transcription → CER Computation → Accept / Reject***

Threshold Tuning: CER = 0.3 accepted < 1% of audio — too strict. Tuned to **0.7** after manually listening to accepted/rejected samples, giving a **~60% acceptance rate.**

Model vs. Language Complexity: faster-whisper is a general-purpose model; Indic scripts have lower baseline accuracy due to complex tokenization, regional dialects, and code-mixing (e.g. English loanwords in Hindi/Tamil).

Acoustic Validity: Manual inspection confirmed files scoring **0.30–0.70 CER** were valid, high-quality speech — Whisper was making spelling errors and **dropping fast phonemes, not flagging bad audio.**

Rejection rate:

- Leniency of **0.7 CER is set**, the model rejects **40%** of the bad audio samples and keeps **60% back.**
- Through manual verification of removed samples it is working perfectly that transcripts don't accurately match for noisy and unclear audio.
- The Whisper model did a **good job in filtering** out bad samples with a decent CER threshold.

Scalability Approach(Bonus):

- **Hybrid Concurrency for Data Ingestion: ProcessPoolExecutor** parses .parquet files + nested **ThreadPoolExecutor** handles I/O-bound audio byte extraction to disk.
- **Acoustic Parallelization – Soft Scoring Phase: ProcessPoolExecutor** runs heavy signal processing (FFTs, RMS energy) across thousands of files simultaneously, **maximizing CPU utilization.**
- **Zero-Idle 3-Stage Whisper Pipeline – Whisper Phase: Eliminates GPU starvation** by running three concurrent stages — I/O prefetching (thread pool reads .flac → RAM), GPU transcription (**RAM → VRAM, never stalled**), and CPU CER computation (**Levenshtein distance** calculated for the previous batch while GPU processes the next).

- **Hardware-Agnostic Scalability:** Auto-detects available resources at runtime — scales from **2-core CPU + integrated GPU (int8, batch size 1)** up to 32-core CPU + high-end GPU like **RTX 3090** (float16, batch size 32), ensuring **high throughput and zero OOM crashes** across any hardware configuration.

Final Insights and Key Inference:

- The Indic dataset consists of lot of low-mid samples which is found using manual listening to sample of audio and found out most **prominent problems** in audio samples.
- Developed a hybrid Soft Score pipeline, applying per-language normalization to highly variable metrics **(VAD, SNR)** and global normalization to structural flaws.
- Set the **rejection threshold** to a inflection point **(0.50),** safely pruning dead air and toxic noise without erasing naturally slower or hissier languages.
- Applying a global Voice Activity (VAD) threshold **of > 0.60** would preserve almost all Tamil data but unfairly **delete over 40%** of Hindi data, proving the absolute necessity of **per-language normalization.**
- Shifting the baseline Soft Score rejection threshold from **0.40** to the CDF's natural quality cliff at **0.50** corrected an artificially **low 2.2%** rejection rate to a more accurate **~5% rejection rate.**
- Calibrated the **Character Error Rate (CER)** threshold to **0.70**, purging catastrophic transcript mismatches while forgiving Whisper's baseline struggles with complex Indic dialects.
- **The Final Impact:** The two-phase pipeline guarantees a sanitized, densely voiced, and accurately transcribed dataset that is strictly balanced and good for ASR training.